INCH POUND

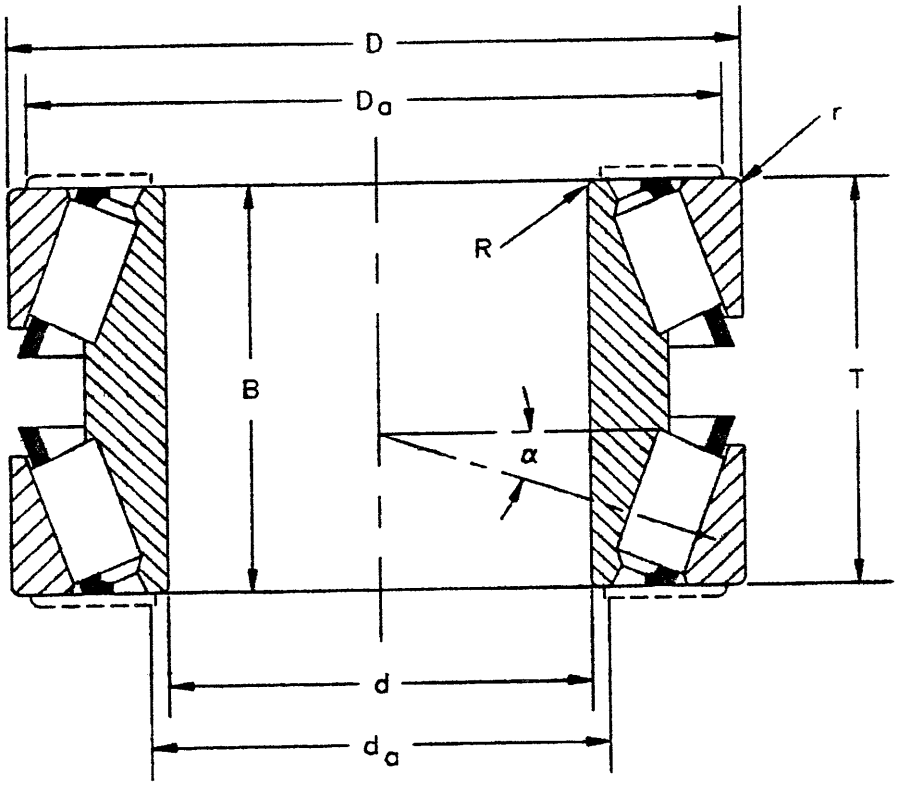
FF-B-187/5
10 SEPTEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, ROLLER, TAPERED, DOUBLE ROW, ONE DOUBLE CONE, TWO SINGLE CUPS. (TYPES TDI AND TDIS)

This specification is approved by the Administrator, Federal Supply Service, General Services Administration, for the use of all federal Agencies.

The complete requirements for procuring the bearings described herein, shell consist of this document and the latest issue of FF-B-187/GEN.



BEARING CONFIGURATION

FSC 3110

AMSC N/A

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
01	17116D-17244	1.1875	2.4409	1.3306	1.3750	0.03	0.06	1.40	2.13	1.53	12000
02 *	14126D-14276	1.2500	2.7170	1.5632	1.5426	0.06	0.05	1.57	2.36	1.53	15100
03	19145D-19268	1.4375	2.6875	1.4492	1.5000	0.03	0.06	1.67	2.40	1.31	14200
04	19145D-19281	1.4375	2.8125	1.4492	1.5000	0.03	0.04	1.67	2.48	1.31	14200
05	19145D-19283	1.4375	2.8346	1.5382	1.5000	0.03	0.06	1.67	2.48	1.31	14200
06	358D-354A	1.6875	3.3465	1.9790	2.0620	0.06	0.05	2.01	3.03	1.91	22800
07	358D-352	1.6875	3.5480	2.1650	2.0620	0.06	0.09	2.01	3.07	1.91	22800
08 *	376DE-374	1.8750	3.6718	1.9690	2.0940	0.03	0.05	2.17	3.35	1.37	24400
09	375D-374	2.0000	3.6718	1.9690	2.0940	0.03	0.05	2.24	3.35	1.73	24400
10	399D-394A	2.4375	4.3307	2.1870	2.1870	0.03	0.05	2.76	3.98	1.45	27700
11	399D-394AS	2.4375	4.3307	2.1870	2.1870	0.03	0.13	2.76	3.90	1.45	27700
12	39585D-39520	2.5000	4.4375	2.3750	2.3750	0.03	0.13	2.83	3.98	1.72	46200
13	39585D-39521	2.5000	4.4375	2.3750	2.3750	0.03	0.03	2.83	4.06	1.72	46200
14	78251D-78537	2.5000	5.3750	2.6020	2.5980	0.09	0.13	3.10	4.53	.67	55200
15	78251D-78551	2.5000	5.5130	2.6020	2.5980	0.09	0.09	3.11	4.61	.67	55200
16	LM114848D-LM114811	3.0000	4.5625	2.1250	2.1250	0.03	0.06	3.27	4.21	2.15	34600
17 *	765D-752	3.0000	6.3750	3.9980	4.2500	0.14	0.13	3.75	5.75	1.71	89900
18	496D-492A	3.1875	5.2500	2.3750	2.3440	0.06	0.13	3.58	4.72	1.31	42800
19	496D-493	3.1875	5.3750	2.3750	2.3440	0.06	0.13	3.58	4.80	1.31	42800

* INACTIVE FOR NEW DESIGN AFTER 20 MAR 1978.

1/ PART NUMBERS ARE FOR REFERENCE ONLY. PART NUMBERS REFLECT THE CUP AND CONE NUMBERS USED BY INDUSTRY AND THE AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA).

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TABLE I. DASH NUMBERS AND DIMENSIONS - Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
20	581D-572	3.1875	5.5115	3.1875	3.1549	0.06	0.13	3.62	4.92	1.45	51100
21	L116149DA-L116110	3.2500	4.5625	1.6250	1.9375	0.06	0.06	3.54	4.25	1.90	15000
22	L217845D-L217810	3.3750	4.8750	1.6251	1.7500	0.03	0.06	3.66	4.57	1.77	25900
23	L217849DA-L217810	3.5000	4.8750	1.6251	1.9063	0.06	0.06	3.82	4.57	1.77	25900
24 *	767D-752A	3.5000	6.3750	3.9980	4.2500	0.06	0.13	3.99	5.75	1.71	89700
25 *	865D-854	3.5000	7.5000	4.6250	5.0000	0.38	0.13	4.57	6.69	1.74	125400
26	42362D-42584	3.6250	5.8437	2.2500	2.2812	0.06	0.12	4.06	5.28	1.19	45300
27	LM119348D-LM119311	3.7500	5.3750	2.2500	2.2500	0.03	0.09	4.02	4.96	2.06	38700
28	868D-854	4.0000	7.5000	4.6250	5.0000	0.06	0.13	4.57	6.69	1.74	125400
29 *	945D-932	4.0000	8.3750	5.6250	6.0000	0.41	0.13	5.25	7.36	1.79	142500
30	LM121948D-LM121911	4.2500	6.0000	2.5625	2.5625	0.03	0.13	5.51	4.53	2.07	49500
31	71426D-71750	4.2500	7.5000	3.8750	4.0000	0.06	0.13	4.80	6.73	1.40	100000
32	946D-932	4.2500	8.3750	5.6250	6.0000	0.13	0.13	5.04	7.36	1.79	142500
33	71450D-71750	4.5000	7.5000	3.8750	4.0000	0.06	0.13	5.04	6.73	1.40	100000
34 *	HH224346DD-HH224310	4.5000	8.3750	5.6250	6.0000	0.13	0.13	5.24	7.56	1.79	185600
35	LM124449D-LM124410	4.7500	6.5625	2.8440	2.8440	0.03	0.13	5.04	6.06	1.99	58000
36	M224749D-M224711	4.7500	6.7812	2.6251	2.6875	0.03	0.06	5.08	6.38	1.76	68000
37	M224749D-M224710	4.7500	6.8750	2.6251	2.6875	0.03	0.06	5.08	6.42	1.76	68000
38	48290DW-48220	5.0000	7.1875	3.0000	3.0000	0.06	0.13	5.39	6.61	1.91	74000
39 *	67388D-67322	5.0000	7.7500	3.6250	3.6250	0.06	0.13	5.50	7.09	1.70	97800

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2/ THESE MAXIMUM FILLET RADII SHALL BE CLEARED BY THE BEARING CORNERS.

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TABLE I. DASH NUMBERS AND DIMENSIONS -Continued

DASH NUMBER	PART NUMBER 1/ CONE-CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
40 *	EE153053D-153100	5.0000	10.0000	6.3750	6.7500	0.13	0.25	5.88	8.86	1.82	205200
41	745100-74845	5.1183	8.4636	4.8750	4.8750	0.06	0.13	5.75	7.72	1.20	98800
42	745100-74850	5.1183	8.5000	4.8750	4.8750	0.06	0.13	5.75	7.72	1.20	98800
43	673900-67322	5.2500	7.7500	3.6250	3.6250	0.06	0.13	5.71	7.09	1.70	97800
44	48393D-48320	5.3750	7.5000	3.0625	3.0625	0.06	0.13	5.67	6.97	1.82	75000
45	H228649D-H228610	5.3750	8.8750	4.7500	4.7500	0.06	0.13	5.98	7.99	1.76	177500
46	486800-48620	5.5000	7.8750	3.0625	2.9688	0.03	0.13	5.91	7.28	1.74	74600
47	74555D-74845	5.5000	8.4636	3.7500	4.1875	0.06	0.13	6.06	7.72	1.20	99000
48	74555D-74850	5.5000	8.5000	3.7500	4.1875	0.06	0.13	6.06	7.72	1.20	99000
49	M231649D-M231610	6.0000	8.7500	3.3125	3.3125	0.06	0.06	6.50	8.15	1.76	94300
50 *	99603D-99100	6.0000	10.0000	6.2500	6.2500	0.06	0.13	6.69	8.94	1.43	156800
51 *	HH234049D-HH234018	6.0000	12.5000	6.3750	6.7500	0.38	0.27	7.15	11.02	1.79	333400
52	4509000-451212	6.0000	12.1250	6.3750	6.7500	0.38	0.27	7.38	10.59	1.79	252300
53	467900-46720	6.5000	8.8750	3.1250	3.0000	0.03	0.13	6.89	8.23	1.52	78200
54	46791DW-46720	6.5000	8.8750	3.1250	3.1250	0.03	0.13	6.89	8.23	1.52	78200
55	677900-67720	7.0000	9.7500	3.5625	3.5625	0.06	0.13	7.48	9.02	1.33	66800
56	82681D-82620	7.0000	11.0000	4.4374	4.4375	0.06	0.13	7.68	9.88	1.11	90800
57 *	EE91700D-91112	7.0000	11.2500	4.1874	4.1875	0.06	0.13	7.64	10.28	1.35	134000
58	94706D-91113	7.0000	11.3750	4.8750	4.8750	0.06	0.13	7.68	10.20	1.35	174200
59	HM237546D-HM237510	7.0000	11.3750	4.8750	4.8750	0.06	0.13	7.64	10.47	1.83	211800

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TABLE I. DASH NUMBERS AND DIMENSIONS –Continued

DASH NUMBER	PART NUMBER 1/ CONE–CUP	d BORE	D OUTSIDE DIA	T BEARING WIDTH OVER CUPS	B CONE WIDTH	R 2/ MAX SHAFT FILLET RADIUS	r 2/ MAX HOUSING FILLET RADIUS	d_a RECOMMENDED SHOULDER DIAMETER SHAFT	D_a RECOMMENDED SHOULDER DIAMETER HOUSING	K FACTOR	BASIC DYNAMIC LOAD RATINGS (LB) TWO ROW RADIAL
60 *	HM237546DD–HM237510	7.0000	11.3750	6.2500	6.2500	0.06	0.13	7.63	10.47	1.83	211800
61	HM237546D–HM237513	7.0000	11.4163	4.8750	4.8750	0.06	0.12	7.64	10.51	1.83	211800
62	EE280700D–281200	7.0000	12.0000	4.3086	4.5000	0.13	0.13	7.87	10.98	1.62	174800
63	M238849DW–M238810	7.3750	10.6250	4.0000	4.0000	0.06	0.13	7.87	9.84	1.76	151000
64	H239649D–H239610	7.3750	12.5970	6.6250	6.3750	0.13	0.19	8.23	11.54	1.83	303000
65	H239649D–H239612	7.3750	12.6250	6.6250	6.3750	0.13	0.19	8.23	11.54	1.83	303000
66	67885DW–67820	7.5000	10.5000	3.5625	3.5313	0.06	0.13	8.03	9.69	1.22	115900
67	EE420750D–421437	7.5000	14.3720	6.2500	6.0000	0.13	0.13	8.70	12.95	1.45	337000
68	93800D–93125	8.0000	12.5000	4.8750	4.8750	0.06	0.13	8.74	11.26	1.12	200400
69	93801D–93125	8.0000	12.5000	5.2500	5.2500	0.25	0.13	8.93	11.26	1.12	200400
70	EE420800D–421437	8.0000	14.3720	6.2500	6.0000	0.13	0.13	9.06	12.95	1.45	320200
71	EE420800D–421450	8.0000	14.5000	6.2500	6.0000	0.13	0.13	9.06	13.03	1.45	320200

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REQUIREMENTS:

1. Material: Cones, cups, and rollers: bearing quality - carburizing grade alloy or through-hardening grade alloy steel in accordance with ASTM-A295, ASTM-A534, or ASTM-A535. Cage: carbon steel piece stamped).

2. Hardness: Cones, cups and rollers: 58 to 64HRC.

3. Tolerances: Standard grade (AFBMA Class 4) (allowable tolerances are in 0.0001 inch).

Cone bore (d)			
Size range B		Tolerance	
Over	Incl.	Plus	Minus
0 3.0000	3.0000 8.0000	5 10	0 0

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000 12.0000	12.0000 24.0000	10 20	0 0

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0 4.0000	4.0000 12.0000	160 280	0 200

Assembled bearing maximum radial runout		
Cup O.D. (D)		Tolerance
Over	Incl.	
0	24.0	20

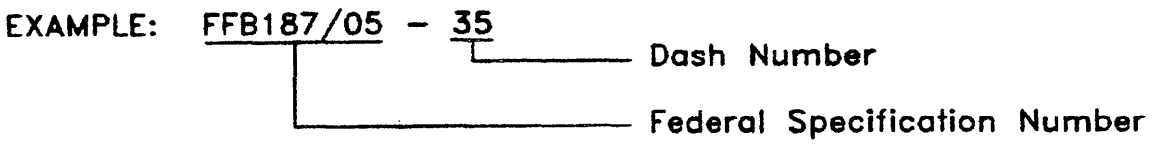
4. Operating temperature: Recommended operating temperature not to exceed 230° F.

5. Lubrication: Bearings shall be furnished without lubrication. All bearings shall be preserved in accordance with MIL-P-197.

6. Control angle: All bearings except dash numbers 14 and 15 are normal angle bearings having a contact angle (a) between 10 and 19 degrees. Bearing dash numbers 14 and 15 are a steep angle bearings having a contact angle (cc) between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

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7. Part number: The part number consist of the letters FFB, the basic federal specification number, the Specification sheet number and a dash number from table I.



8. For design feature purposes, this standard takes precedence over procurement documents referenced herein.

9. Referenced documents shall be of the issue in effect on the date of invitation for bids, or request for proposal.

NOTE: Inch units can be converted to soft metric units as follows:

Inch	X	25.4	=	millimeters

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INTERCHANGEABILITY TABLE

MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS	MILITARY STD PART NUMBERS	SUBSTITUTIVE PART NUMBERS
MS19085-1	FFB187/05-02	MS19085-24	FFB187/05-18
MS19085-2	FFB187/05-08	MS19085-25	FFB187/05-20
MS19085-3	FFB187/05-12	MS19085-26	FFB187/03-26
MS19085-4	FFB187/05-17	MS19085-27	FFB187/05-31
MS19085-5	FFB187/05-24	MS19085-28	FFB187/05-37
MS19085-6	FFB187/05-25	MS19085-29	FFB187/05-38
MS19085-7	FFB187/05-28	MS19085-30	FFB187/05-42
MS19085-8	FFB187/05-29	MS19085-31	FFB187/05-43
MS19085-9	FFB187/05-33	MS19085-32	FFB187/05-44
MS19085-10	FFB187/05-34	MS19085-33	FFB187/05-45
MS19085-11	FFB187/05-39	MS19085-34	FFB187/05-46
MS19085-12	FFB187/05-40	MS19085-35	FFB187/05-53
MS19085-13	FFB187/05-50	MS19085-36	FFB187/05-55
MS19085-14	FFB187/05-51	MS19085-37	FFB187/05-58
MS19085-15	FFB187/05-57	MS19085-38	FFB187/05-65
MS19085-16	FFB187/05-60	MS19085-39	FFB187/05-67
MS19085-17	FFB187/05-69	-	-
MS19085-18	FFB187/05-71	-	-
MS19085-19	FFB187/05-01	-	-
MS19085-20	FFB187/05-03	-	-
MS19085-21	FFB187/05-9	-	-
MS19085-22	FFB187/05-10	-	-
MS19085-23	FFB187/05-15	-	-

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